

Development of a Touchable VR Planetarium for Both Sighted and Visually Impaired People

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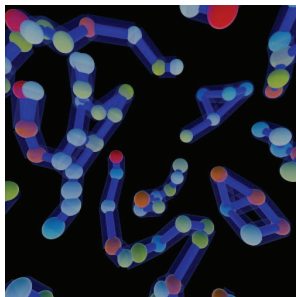


Figure 1: The displayed celestial sphere.

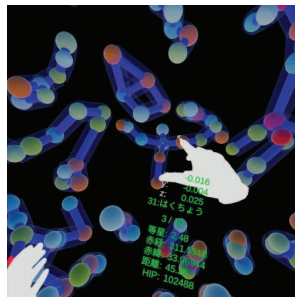


Figure 2: "Touching" Cygnus.

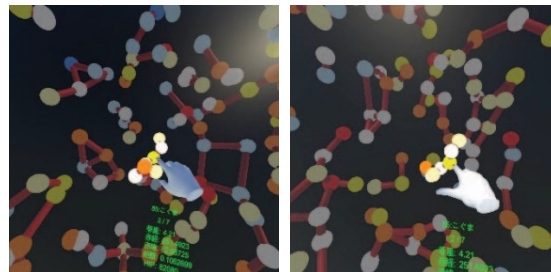


Figure 3: View of the selected Ursa Minor from left to right; The distance from the earth is reflected and presented.

ABSTRACT

The authors proposed and developed a "touchable" VR planetarium. The user wears a VR headset and "touches" the stars with the controllers. Because we can't touch the stars in reality, this application provides the users with additional value and experience of the planetarium. As this feature is valuable for visually impaired people to experience the starry sky, the authors also implemented the functions that help it. In the trial use by visually impaired people, they experienced the starry sky with the support functions and evaluated the VR planetarium as a valuable application.

Keywords: Planetarium, Visually-impaired, Tactile display

Index Terms: Human-centered computing — Human computer interaction (HCI) — Interaction paradigms — Virtual reality

1 INTRODUCTION

VR planetarium applications have been developed. The user wears a VR headset and watches stars with the audio guide. The authors propose letting the user "touch" the stars, i.e., tactile feedback, in the VR planetarium. Though we can't touch the stars in reality, the "touch" feature will make the VR planetarium more valuable and enjoyable. In addition, "touching" the stars will let the visually-impaired feel the stars. In this paper, the authors report a "touchable" VR planetarium application that both sighted and visually impaired people will enjoy based on the concept of universal design.

As related works, a virtual star chart application for smartphones is released. When the user holds the device up to the sky, the star chart which matches the sky is displayed. Planetarium applications for VR headset environments are also released. As for the tactile presentation of the starry sky and constellations, the following studies for the visually impaired are found: a star chart with braille and emboss-printed stars [1], celestial models that allow users to feel

the surface topography [2], and 3D-printed plates of constellations [3]. Most of them are provided as static objects.

In this paper, the authors report the development of a VR planetarium with which the user can "touch" the stars with the controllers. The user can not only "touch" the stars on the celestial sphere but also feel the star's attributes and the distance to the stars. The authors implemented these features with a universal design in mind.

2 DEVELOPMENT OF A TOUCHABLE VR PLANETARIUM

The implemented VR planetarium employs a VR headset (Meta Quest 2) and its controllers, which will be a typical environment. The controllers can present vibrations with various intensities. They represent the sensation of *touching* the stars. The development environment is Unity on Windows.

2.1 Basic Functions as a VR Planetarium

Displaying stars and constellation lines on the celestial sphere is the fundamental function of the VR planetarium (Figure 1). The radius of the celestial sphere is about the length of the arm. Data of stars are from Astro Commons [4]. Stars are colored according to their spectra and displayed as larger spheres at higher apparent magnitudes. The colors and distances from the earth of the stars (described in 2.2) were based on the Hipparcos Catalogue [5].

2.2 Select a Constellation

The most remarkable feature of this VR planetarium is that the user can "touch" the stars. The user can "touch" the stars on the celestial sphere. The "touch" sensation is represented by the vibration of the *right* controller. For example, in Figure 2, the user "touches" Cygnus. Still more, when the user selects a constellation, a detailed touch experience (observation) is provided.

First, the user selects a constellation from the celestial sphere described in 2.1 and "fetches" it by pointing or "touching" it by pressing the controller's button. The constellation's name is read out when it is pointed and when it is "fetched." Alternatively, the user can select the constellation with the controller's stick. At this time, the constellation's name is read out. When the desired constellation is selected, it is "fetched" by pressing the button. The "fetched" constellation is sometimes out of the user's view. In such cases, its name is heard from the direction of it with three-dimensional (3D) sound (this is also one of the features for visually impaired people).

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2.3 “Touch” the Stars in the Constellation

The “fetched” constellation is placed within reach. Here, the stars consisting of the constellation are not arranged in a plain but in a three-dimensional presentation reflecting their distances (deformed) from the earth. For example, when the user “fetches” Ursa Minor and looks around it to the left and right, s/he can see how far the stars which consist of it are (Figure 3).

The user can “touch” the stars in the “fetched” constellation. S/he reaches out to “touch” farther stars. When s/he “touches” a star, the *right* controller vibrates according to its apparent magnitude (the brighter, the stronger). These are the remarkable features of this VR planetarium. When the user touches the constellation line, the *left* controller vibrates. It is a guide for visually impaired people.

Information about the “touched” star is also displayed: the constellation name it belongs to, apparent magnitude, position on the celestial sphere, distance, and so on.

3 UNIVERSAL DESIGN FOR THE VISUALLY-IMPAIRED

Visually impaired users can observe the starry sky through the “touch” feature of the VR planetarium. In order to improve this feature, several functions to assist visually impaired users have been implemented. They include a selection of constellations with the stick (Section 2.2) and a vibration guide of constellation lines (Section 2.3). These features should also be useful for sighted users.

3.1 Audio Feedback

Sound and voice feedback is provided for each operation. Though tactile feedback is implemented, audio feedback for operations is also important (whether the user is visually impaired or sighted).

3.2 Audio Guides

The selected constellation’s name is read out whenever by pressing the controller’s button. When a constellation is “fetched,” the number of stars belonging to it is read out to help the user grasp its form.

Each star in the constellation has its own number. When a star is “touched,” this number is read out. In this way, the user knows which star in the constellation s/he is “touching.” When s/he touches all the stars in the constellation, a jingle is played.

When the user “touches” a star and presses the controller’s button, the star’s color is read out.

3.3 Guides Using Three-dimensional Sound

The 3D sound indicates the position of the selected constellation (described in 2.2). In addition, when the user “touches” the stars in the constellation and presses the controller’s button to ask for assistance, the number of the star next to the last “touched” star is read out from the direction of the star with a 3D sound.

3.4 Guides Using Vibration

Vibration represents the “touch” sensation of the stars and a guide to the stars (as described in 2.3). As for the latter, when the user is “touching” the constellation lines, the *left* controller vibrates. When the user’s hand is near the constellation lines, the *left* controller slightly vibrates. These vibrations lead the user to the stars.

3.5 All Reset

Whenever the user long-presses the controller’s home button, the application is reset and initialized. Visually impaired people tend to lose the state of the application because it is hard to see the state. A reset function like this is necessary for them.

4 TRIAL BY THE VISUALLY-IMPAIRED

Three visually impaired people tried the VR planetarium application. They were totally blind. They all had seen the stars before their

sights got worse. One of them was familiar with astronomy.

In the trials, they sat on the chair and wore the VR headset. They experienced the VR planetarium with the authors’ explanation. Each of them experienced about an hour. Only when the participants lost the application’s state, the authors helped their operation.

4.1 How They Experienced

First, as they reached out and explored their surroundings, they were able to sense that they were “touching” the stars through sound and vibration. They found that sound pitch and vibration intensity represented the star’s brightness. They were able to explore and grasp the stars on the celestial sphere.

They were able to select the constellation and “fetch” it using the voice guide. One of them preferred to select the constellation with the controller’s stick and hear where the constellation was.

They were able to explore the stars of the “fetched” constellation, relying on each star’s number. Vibration on the constellation lines helped the explore. They also checked the “touched” star’s color.

Overall, they were able to experience the VR planetarium as the authors expected with the authors’ guide.

4.2 Their Comments

They all said, “I have never had such an interesting experience.” Considering their little VR experience, this VR planetarium is valuable VR content for the visually impaired. One of them said, “I could imagine what the starry sky looks like.” This comment indicates it will be useful for those who have not seen the starry sky. They said that it was preferable that the equipment was inexpensive (different from special equipment for visually impaired people).

The participant who was familiar with astronomy said, “It is good that all constellations are contained.” This mention reflects that most astronomy content for visually impaired people doesn’t contain all constellations. The VR planetarium’s advantage is that it is implemented digitally and that it is flexible.

They requested a tutorial to learn the system and content that introduces the popular constellations for beginners (utilizing flexibility). By implementing them, the system will become more valuable.

5 CONCLUSION AND FUTURE WORKS

In this paper, the authors introduced a “touchable” VR planetarium for both sighted and visually impaired people. Its remarkable feature is that the user can “touch” the stars. The authors thought it was valuable for visually impaired people to experience the starry sky and implemented the assisting functions. The authors had three visually impaired people try the VR planetarium. They were able to experience it as the authors expected and said that it was interesting.

One of the future works is the improvement of operation. A more comfortable experience should be provided. Content requested in the trial use is to be made. Besides, as there are various types of visual impairment other than total blindness, considering them is also an essential issue to increase the value of the VR planetarium.

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